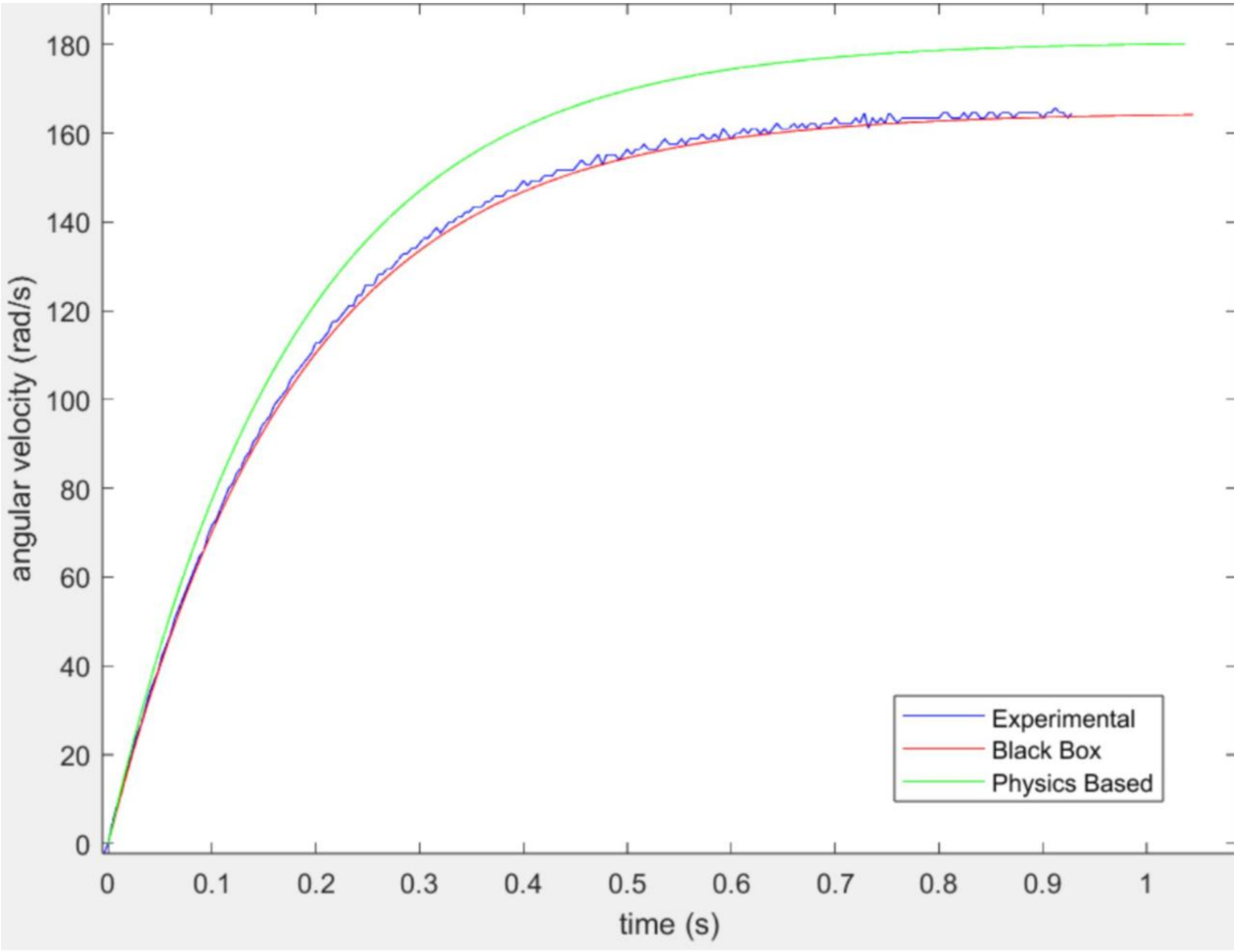
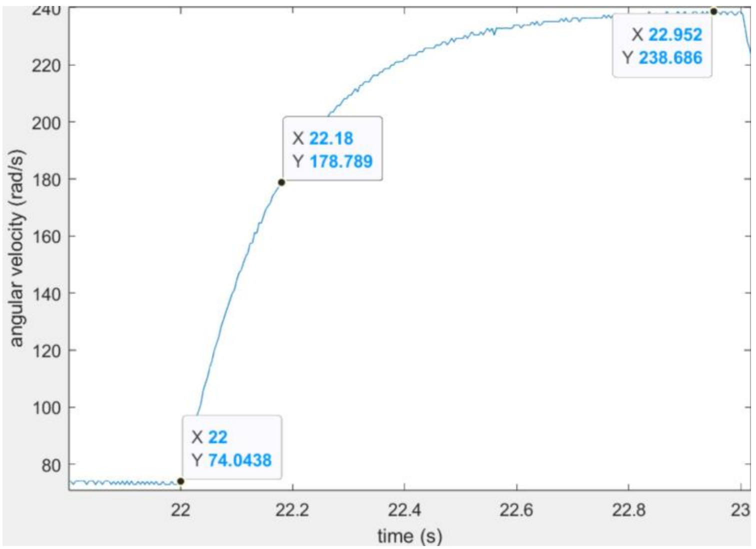
Sample Results

Lab 2: Parameter ID and step response data





Tables and Calculations for R and K_b: Take a screenshot of the two tables from the Word document used in InClass 2 that were used in the calculation for R and K_b here:

Screenshot of Inclass table with voltages and current for R calculation

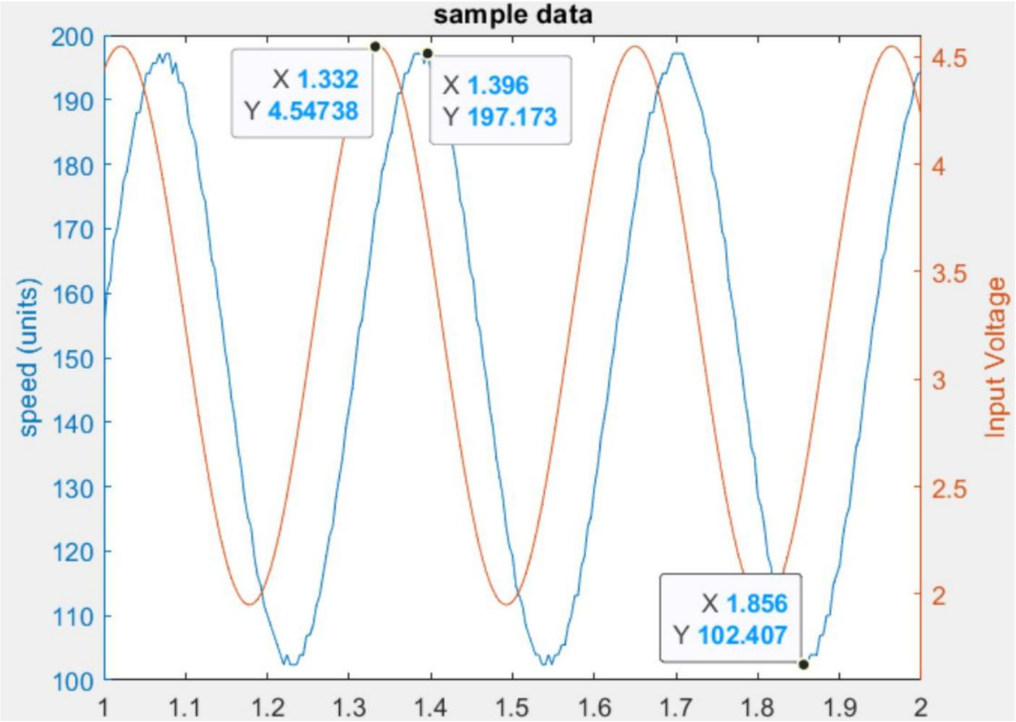
Specified voltage: V_s [V]	Measured voltage: V_m [V]	Measured current: I_m [Amp]	Resistance: R_m (Ω)
0.8	0.72	-0.1	7.2
1.6	1.45	-0.19	7.63
2.7	2.40	-0.333	7.21
3.79	3.15	-0.425	7.41
	Average Resistance: R_{avg} (Ω)		7.36

d. Measure voltage and current, referring to 'Voltage, Current, Measurements' for

Screenshot of Inclass table with voltage, current and velocity for Kb calculation

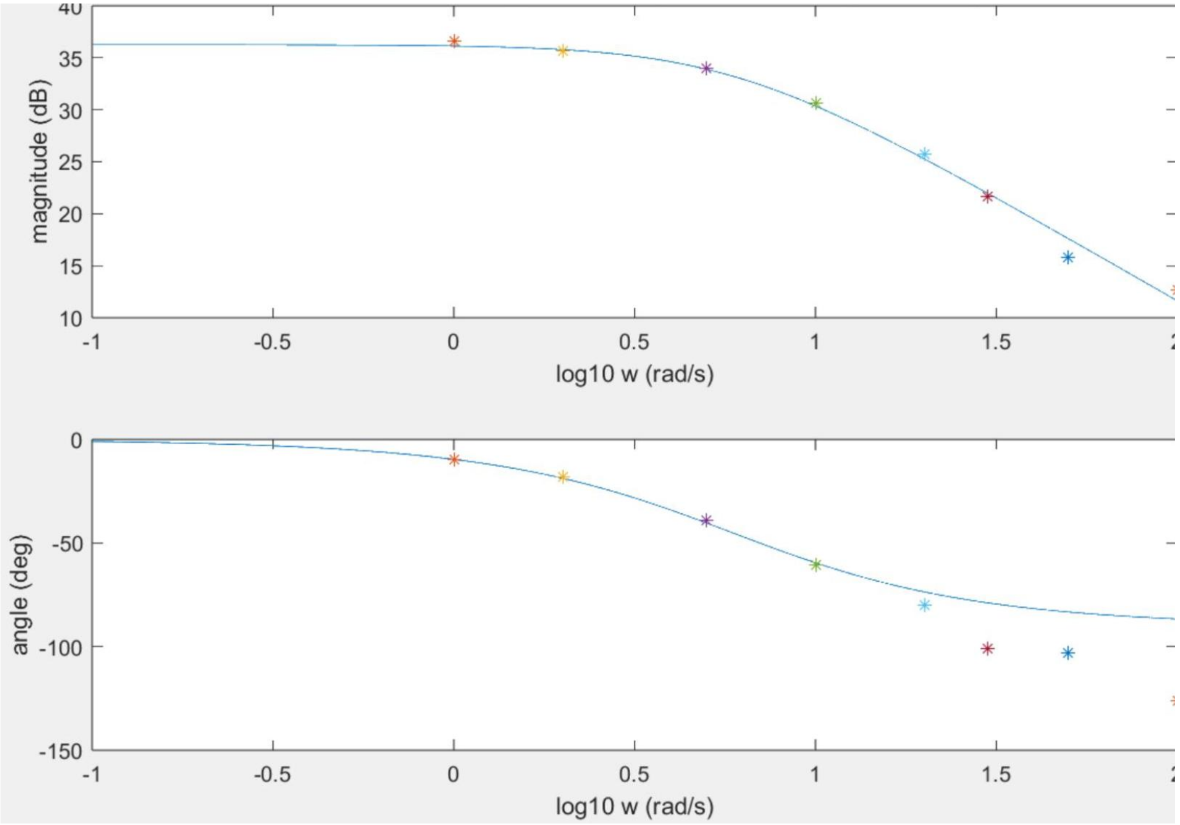
specified voltage: Vs [V]	measured voltage: V_m [V]	measured current: I_m [Amp]	measured speed: ω_m [rad/s]	k_b [V.s/rad]
0.8	0.74	-0.07	16	0.01405
2	1.93	-0.076	88	0.01558
3.5	3.43	-0.085	180	0.01558
4.5	4.42	-0.092	239	0.01566
	Average Back EMF-Constant: $k_{b,avg}$ [V.s/rad]			0.01522

Lab 3: Frequency Response

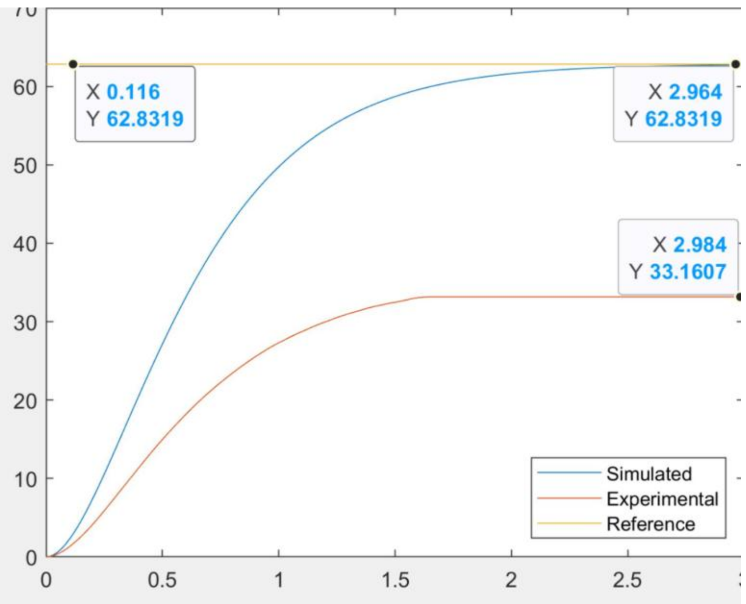


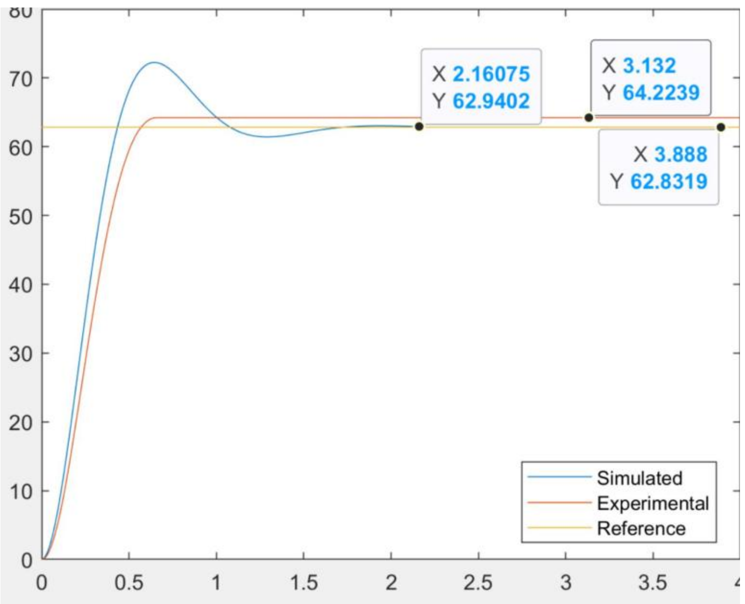
1.3 Summarize the full set of data in the table below.

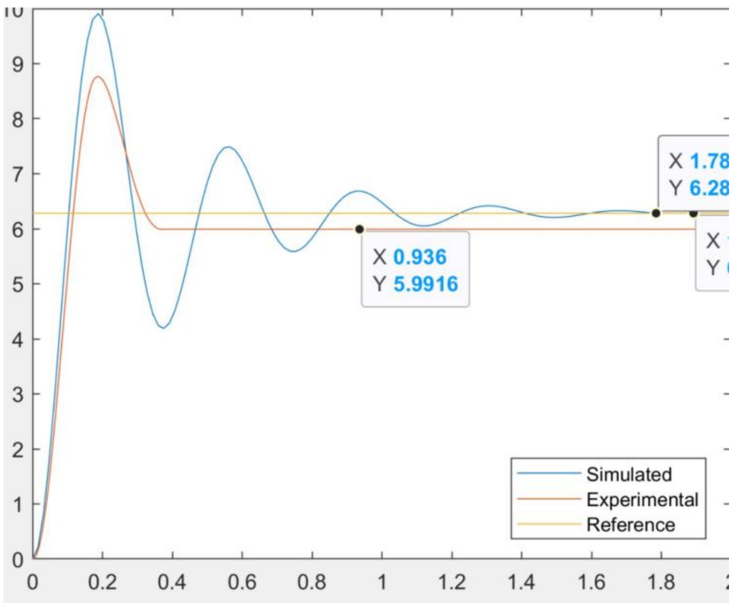
Frequency (Rad/s)	**Magnitude (Rad/s/V)**	**Magnitude (dB)**	**Time shift (s)**	**Phase *(Degrees)***
1	67.71	36.61	0.168	-9.68
2	60.361	35.615	0.158	-18.105
5	50.165	34.008	0.136	-38.961
10	34.10	30.665	0.106	-60.734
20	19.261	25.67	0.07	-80.214
30	12.167	21.704	0.056	-100.78
50	6.175	15.813	0.036	-103.132
100	4.314	12.698	0.022	-126.051



Lab 5: Proportional Position Control







Lab 6: speed Control